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AirBnb Pricing Dynamics in New York City Across COVID Eras

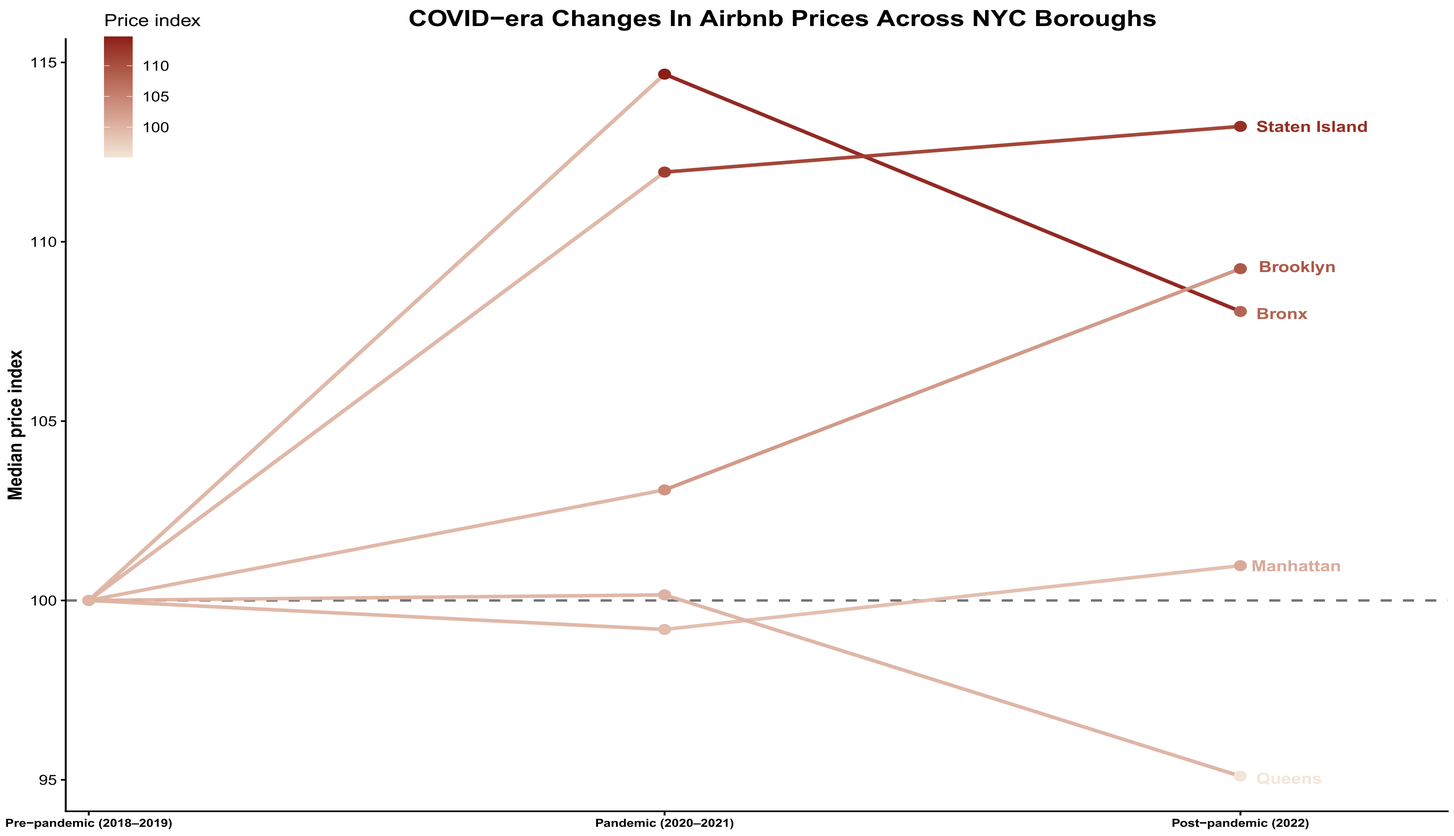
Story

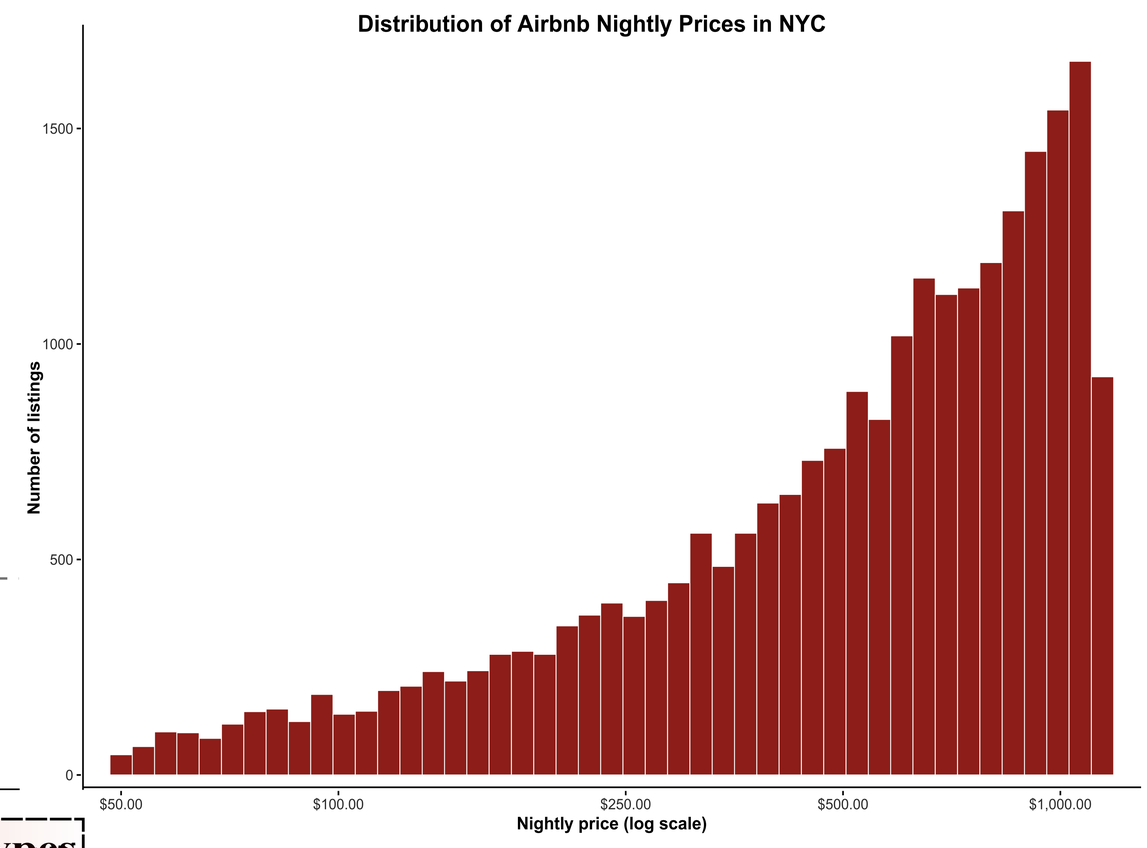
This poster quantifies how Airbnb prices in New York City shifted across COVID eras by room type, neighborhood, price tier, and distribution shape. Using normalized price indices, distribution shifts, and listing flows, it reveals uneven recovery patterns that reshaped the city's short-term rental market .

Motivation

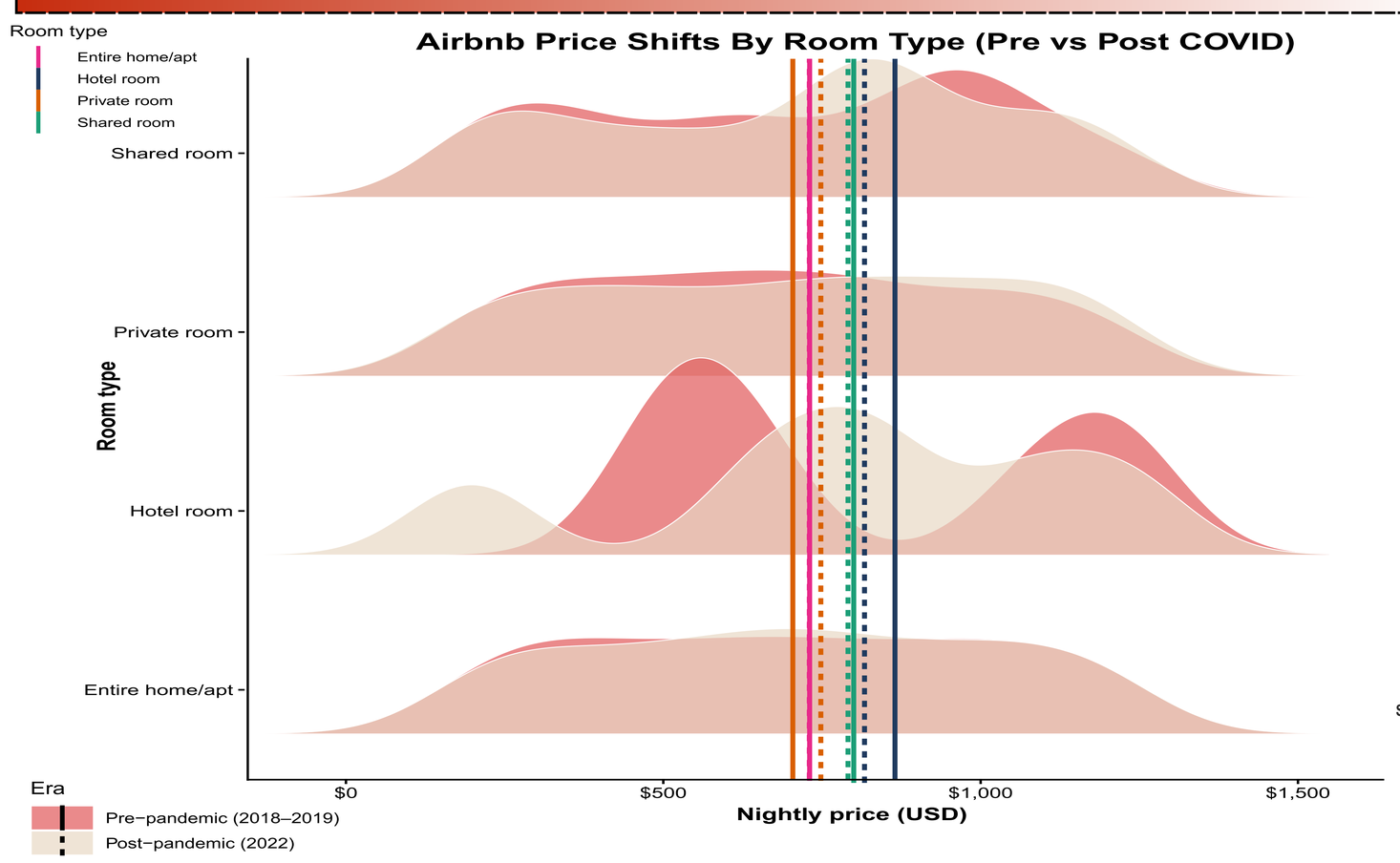
Travelers and hosts encountered a more unpredictable market as Airbnb pricing in New York City changed during and afterCOVID-19. While hosts, city planners, and researchers require clarity on how pricing patterns varied among room types, areas, and price tiers, travelers want to know where costs changed.

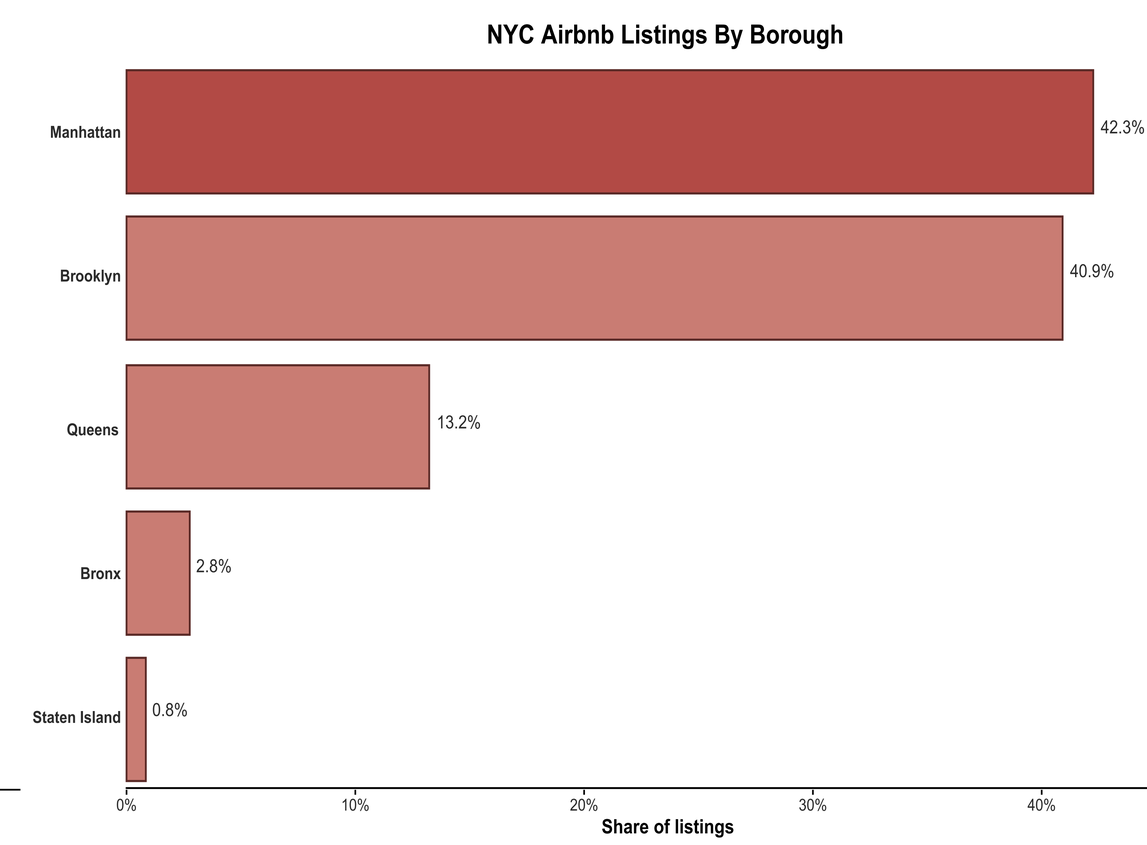
How did Airbnb prices change across NYC boroughs during COVID?



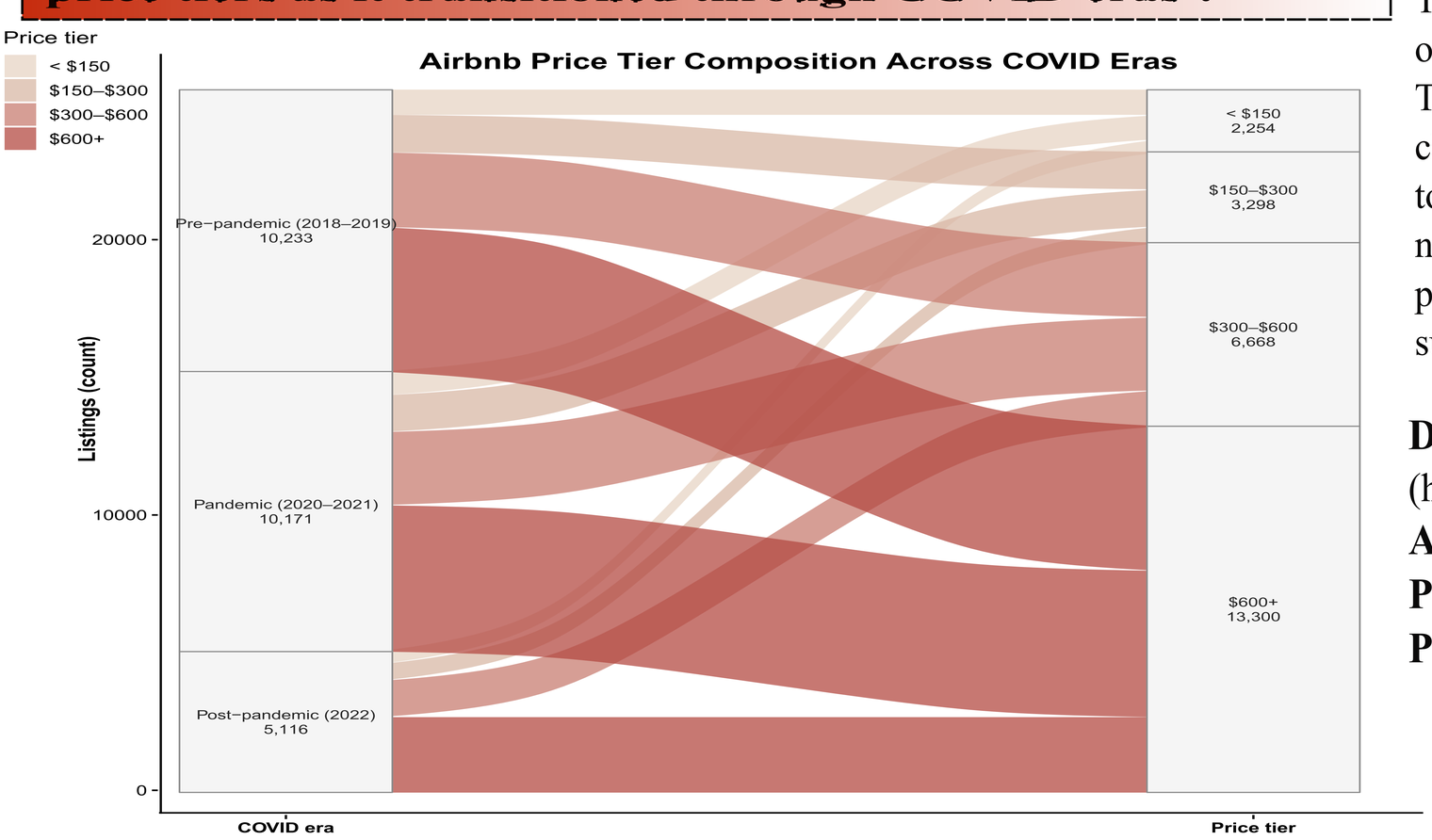


How did the distribution of nightly prices shift across different room types from pre-pandemic to post-pandemic periods?





Did the Airbnb market structurally shift toward higher price tiers as it transitioned through COVID eras ?



Data Description

This analysis uses the New York City Airbnb Open Dataset, originally containing 102,599 listings and 26 variables. The data was subset to 25,520 listings from 2018–2022 and cleaned by removing missing construction years, restricting listings to NYC boroughs, standardizing price fields, and excluding non-positive prices. Listings were grouped into pre-pandemic, pandemic, and post-pandemic eras to support comparative price analysis.

Data sourced - Kaggle (https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata)
Analysis and visualization - R
Packages - dplyr, ggplot2, ggridges, ggalluvial, ggpubr, scales
Poster design - Inkscape